

MUGILAN SARAVANA PERUMAL

Electrical & Electronics Engineering Student | Power Electronics | MATLAB & Simulink

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PROFESSIONAL SUMMARY

Electrical and Electronics Engineering student with hands-on simulation experience in power electronics, DC-DC converters, and closed-loop control systems using MATLAB/Simulink and LTspice. Growing interest in electric vehicle technology, embedded systems, and sustainable energy, backed by practical project work and structured technical training. Combines core electrical engineering fundamentals with programming and web development skills to approach problems from both hardware and software perspectives.

EDUCATION

Sri Manakula Vinayagar Engineering College 2025 – 2029

Bachelor of Technology (B.Tech), Electrical and Electronics Engineering

Amalorpavam Higher Secondary School 2011 – 2025

High School

TECHNICAL SKILLS

Power Electronics & Simulation: Power Electronics, DC-DC Converters, Buck Converter, Boost Converter, PWM, MATLAB, Simulink, LTspice, Proteus

Control Systems: Closed-Loop Control, PI Controller, DC Motor Speed Control, Feedback Control, Dynamic System Simulation

Programming: C, Python

CAD / Design: Autodesk Fusion 360

Front-End Development: HTML, CSS, JavaScript

Database: Microsoft SQL Server (MSSQL)

PROJECTS

Closed-Loop Speed Control of a DC Motor using a DC-DC Buck Converter MATLAB | Simulink | PWM | PI Controller

- Developed a closed-loop DC motor drive simulation using a DC-DC buck converter in MATLAB/Simulink.
- Implemented speed feedback and a PI controller to dynamically adjust PWM duty cycle.
- Evaluated system behavior under reference speed changes and step torque-load disturbances.
- Analyzed voltage ripple, current dynamics, speed regulation, and settling behavior.

Smart Wireless Power Transfer System for EV Charging — IDEA Lab EV Technology | Inductive Power Transfer

- Developed a concept for wireless EV charging using inductive power transfer technology.
- Explored transmitter and receiver coil operation for contactless energy transfer.
- Proposed features including vehicle-position detection, charging-status indication, voltage/current monitoring, and overcurrent protection.
- Identified key measurement parameters, including voltage, current, power, charging status, and transfer efficiency.

Buck Converter Simulation LTspice | PWM | DC-DC Conversion

- Simulated a step-down DC-DC buck converter using LTspice with a 12 V input and 25 kHz switching frequency.
- Modeled the circuit using an IRFZ44N MOSFET, 1N5819 Schottky diode, 143 μ H inductor, 200 μ F capacitor, and 10 Ω load.
- Investigated unexpected output waveform behavior by checking component models, switching operation, and MOSFET gate drive.
- Analyzed transient and steady-state behavior, strengthening understanding of PWM-based voltage regulation and converter verification.

Boost Converter Simulation LTspice | PWM | DC-DC Conversion

- Simulated a boost converter stepping up a 5 V input to approximately 10 V output using LTspice.
- Configured a 100 kHz switching frequency with a 50% duty cycle.
- Built the circuit using a 100 μ H inductor and 100 μ F capacitor.
- Strengthened understanding of PWM switching, inductor energy storage, and DC-DC power conversion.

CERTIFICATIONS

Learn Fusion for CAD in 90 Minutes — Autodesk August 2026

Certification in Fusion 360

MATLAB Certified — MathWorks February 2026

Simulink Onramp Certificate

Recognition of Participation — Skill Development Awareness Program 29 June 2026

TRAINING / TECHNICAL EXPOSURE

- 5-Day EV Training Program — TVS Training & Services

CAREER INTERESTS

Power Electronics • Electric Vehicle Systems • Renewable Energy • Industrial Drives • Control Systems • Embedded Systems